CSHL BIOLOGICAL DATA SCIENCE · NOVEMBER 4–7, 2026

Orbit and Galaxy User-Defined Tools for Reproducible, Auditable Computational Biology

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Five published genomics papers, reanalyzed end-to-end from raw data with an AI agent (**Orbit**) orchestrating Galaxy and its **User-Defined Tools**—and an honest account of what matched, what didn't, and why.

Learn about Orbit
galaxyproject.org/agents/orbit — Auditable agentic analysis on Galaxy.

A Abstract

Computational reproducibility in genomics is often undermined not by the standard steps of an analysis—alignment, peak calling, quantification—but by the manuscript-specific "glue" that connects them: custom filtering, integration, and comparison logic that typically lives in untracked local scripts or notebooks tied to a particular machine's software state. We present a workflow that pairs **Orbit**, an AI-enabled analysis environment, with Galaxy's **User-Defined Tools (UDTs)** to keep this custom logic inside Galaxy as biocontainer-packaged jobs running on shared HPC or cloud infrastructure, with every input, parameter, and output captured as Galaxy history provenance. Because each UDT is a versioned container rather than a notebook cell tied to a particular interpreter and library snapshot, these steps stay executable and reproducible long after the original analysis, avoiding the software rot that erodes notebook-based workflows as dependencies drift.

We apply this workflow to reproduce published papers directly from raw sequencing data. Reanalyzing Chung et al. (2014), a ChIP-seq/RNA-seq study defining a 97-gene SrbA regulon in *Aspergillus fumigatus*, Orbit coordinated reference and workflow discovery through BRC-Analytics and ran eight custom UDTs for RNA-seq contrasts, ChIP/RNA integration, and paper-concordance scoring, surfacing an input-control depth imbalance that had distorted peak calling and recovering **89 of 97 published targets outright**—with all but one of the remaining eight traced to a decade of annotation coordinate drift rather than lost ChIP-seq signal. Reanalyzing Toenhake et al. (2018), an ATAC-seq/RNA-seq study in *Plasmodium falciparum*, UDTs handled chromosome renaming in GTF, alignment filtering, and peak-to-gene integration, recovering **94–96%** of the paper's published correlated gene–peak pairs against the authors' deposited peak universe. Routing custom analysis through UDTs—including logic authored and orchestrated by Orbit itself—converts manuscript-specific glue into shareable, containerized Galaxy histories with full job-level provenance, showing that AI-assisted analysis can be at least as inspectable, reproducible, and long-lived as one written entirely by hand.

B How closely did five reanalyses match the published results?

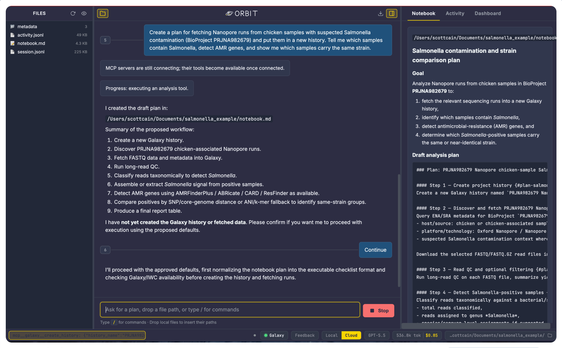
Each bar is that project's own primary match metric — started from raw public reads on Orbit + Galaxy (none copied processed files).



Where a project's own report also flagged a gap (e.g., a 4× peak-count mismatch in Ruiz), the corresponding panel below explains why.

5	67+	3	48+
published papers reanalyzed — 5 organisms, 4 assay types	custom Galaxy tools built where none existed	real Galaxy / IWC bugs found and fixed as a side effect	dated git commits forming an auditable project record

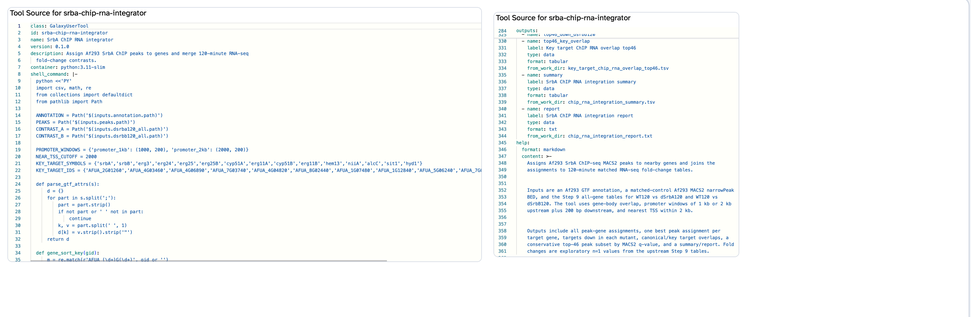
C How a session runs



Orbit in an actual session — project files, chat/plan, and executable `notebook.md` panes.

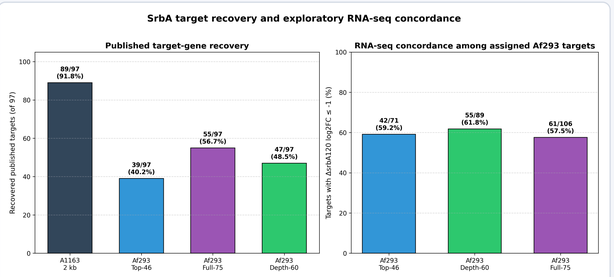


Nothing executes, and nothing is written to the record, until the user approves it — twice.



Example UDT (`srba-chip-rna-integrator`): assigns SrbA ChIP-seq peaks to genes and merges them with matched RNA-seq fold-change data in a single containerized Galaxy job, replacing a fragile multi-step manual join.

D *Aspergillus fumigatus* · ChIP-seq — Chung et al. 2014

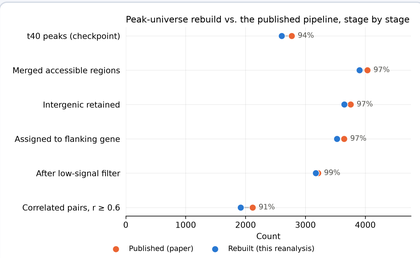


Recovery of 97 published SrbA target genes (left) and RNA-seq concordance among Af293-assigned targets (right) across four peak-calling designs.

- **8 custom Galaxy UDTs** ran the manuscript-specific steps: RNA-seq contrasts, ChIP/RNA integration, and paper-concordance scoring.
- **A depth imbalance between input controls**—easy to miss in a notebook—surfaced as an inspectable Galaxy job and was corrected.
- **89 of 97 published SrbA target genes** recovered outright; all but one of the rest traced to a decade of annotation coordinate drift, not lost signal.

▶ Run → Check → ! Catch → ✓ Fix & record **Caught its own mistake:** a "verified" tool link meant nothing — re-audited by exporting the real UDT definitions.

E *Plasmodium falciparum* · ATAC + RNA-seq — Toenhake et al. 2018

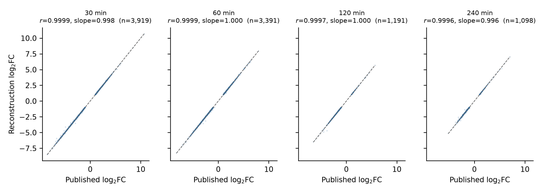


Peak-universe rebuild vs. published pipeline stage by stage — published (orange) vs. reanalysis (blue), with % match at each step.

- **UDTs handled chromosome renaming, alignment filtering, and peak-to-gene integration** across two genome annotation releases, run side by side on purpose.
- **94–96% of correlated gene–peak pairs** recovered using the authors' deposited peaks; **91% recall / 96% precision** rebuilding the peak universe from raw reads alone.
- **An apparent 4× shortfall** turned out to be a smoke-test tool mistaken for a peak caller—not a flaw in the published method.

▶ Run → Check → ! Catch → ✓ Fix & record **Declared the paper "irreproducible,"** then found its own placeholder peak-caller—a named, versioned UDT—and reversed the verdict in 24 hours.

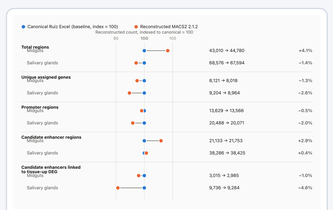
F *A. fumigatus* RNA-seq — Hokken 2018



Reanalysis $\log_2$FC vs. paper's $\log_2$FC at each timepoint (r, slope shown; dashed line $y = x$).

- **DEG counts initially missed the paper by 31–56%** at later timepoints—traced to a modeling choice, not bad data.
- **Fixed to within 2.4%** of published counts; AspGD's retirement replaced with custom GO-enrichment UDTs.
- **Checked its own workflow design against IWC** and overturned its earlier assumption when the evidence didn't support it.

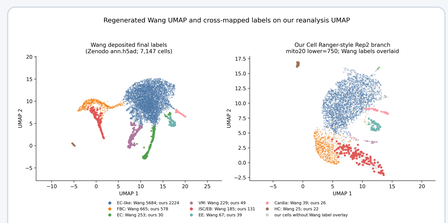
G *An. gambiae* ATAC-seq — Ruiz 2021



Canonical (paper) vs. reconstructed region counts across five regulatory summaries.

- **One UDT failed three different ways** in a single afternoon—each diagnosed, fixed, and logged by version.
- **Peak coordinates reproduced with zero free parameters (92%)**; published peak counts did not—and the report says so plainly.
- **Pre-registered a blind cross-tissue validation test** rather than trust a fitted parameter, reporting "inconclusive" when honest.

H *Ae. aegypti* single-cell — Wang 2024



Left: UMAP from Wang et al.'s deposited labels. Right: reanalysis UMAP with labels cross-mapped onto matching barcodes.

- **Filed 3 real upstream Galaxy / IWC bugs** and contributed 3 reusable Skills—benefits beyond this one paper.
- **Recovered silently erased notebook entries** from the raw activity log two months after a background-process bug.
- **Built a Cell Ranger-style reference** for a mosquito genome Galaxy had no recipe for, fixing non-standard mitochondrial gene naming.

AI-assisted analysis doesn't have to trade away reproducibility. Captured as versioned, auditable Galaxy jobs rather than opaque local scripts, it can be at least as inspectable and durable as analysis written entirely by hand.

Acknowledgements: Work in the authors' groups is supported by National Institutes of Health grants U24HG002371, U24HG006620, U24HG010263, U24CA284167, U24AI183870, and R01GM151683, and by National Science Foundation grants 2419522 and 2138585. Orbit: galaxyproject.org/agents/orbit · UDT beta: udt-signup.galaxyproject.org · Preprint: doi.org/10.64898/2026.10.26.761864. Related CSHL posters: Galaxy UDTs (van den Beek) · Galaxy Workflow Foundry (Chilton) · HighGen (Burhans) · disassembler (Seacord).

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